

Information Theory and Cognition

Problem Set 2: The building blocks of information

Due: Friday 2/13 (before midnight)

1 Axiomatic derivation of Entropy

Let $X \sim p(x)$ be a discrete random variable with values in $\mathcal{X} = \{x_1, \dots, x_n\}$. Denote by $p_i = p(x_i)$, and recall that the entropy of X is defined as

$$H(X) = - \sum_{i=1}^n p_i \log p_i.$$

Shannon derived this function as uniquely satisfying a few axioms that any measure of uncertainty should satisfy. More formally, Shannon showed that if we have a sequence of symmetric functions $H_n(p_1, \dots, p_n)$ such that:

- Normalization: $H_2(\frac{1}{2}, \frac{1}{2}) = 1$,
- Continuity: $H_2(p, 1-p)$ is a continuous function of p , and
- Grouping/additivity: $H_n(p_1, \dots, p_n) = H_{n-1}(p_1 + p_2, \dots, p_n) + (p_1 + p_2)H_2(\frac{p_1}{p_1+p_2}, \frac{p_2}{p_1+p_2})$,

Then $H_n(p_1, \dots, p_n)$ must be the entropy function.

1.1 Show that entropy satisfies these three axioms.

1.2 Proving the uniqueness of entropy is a harder problem. Instead, we will explore alternative functions and see how they fail to satisfy these axioms. We will focus on functions of the form

$$H_{n,f}(p_1, \dots, p_n) = \sum_i p_i f(p_i).$$

Note that entropy is obtained with $f(p) = -\log p$. For each alternative function below, explain whether f could potentially capture surprisal and show which of the axioms $H_{n,f}$ violates.

- (a) $f(p) = \frac{1}{p} - 1$
- (b) $f(p) = n(1-p)$
- (c) $f(p) = \log p$

2 Informational identities

2.1 Let $X_1, \dots, X_n$ be n discrete random variables with joint probability $p(x_1, \dots, x_n)$. Show that

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1)$$

2.2 Let X, Y be discrete random variables with joint distribution $p(x, y)$. Show that:

(a) $I(X; Y) = H(X, Y) - H(X|Y) - H(Y|X)$

(b) $I(X; Y) = H(X) + H(Y) - H(X, Y)$

2.3 Let X, Y, Z be discrete random variables with joint distribution $p(x, y, z)$. The mutual information between Y and (X, Z) is defined as

$$I(Y; X, Z) = \sum p(x, y, z) \log \frac{p(x, y, z)}{p(x, z)p(y)}.$$

The condition mutual information of X and Y given Z is defined as

$$I(X; Y|Z) = \sum p(x, y, z) \log \frac{p(x, y|z)}{p(x|z)p(y|z)}.$$

Show that $I(X; Y|Z) = I(Y; X, Z) - I(Y; Z)$.

2.4 Let X be a discrete random variable and let $g(X)$ be a function of X . Show that $H(X) \geq H(g(X))$.